

NITHISH KUMAR R

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI & Data Science graduate who builds and ships production AI systems — RAG pipelines, LLM applications, and backend services deployed on real cloud infrastructure. Comfortable across the full stack: Python, FastAPI, PostgreSQL/pgvector, Docker, and Google Cloud Run. Focused on turning generative AI concepts into working, reliable software.

TECHNICAL SKILLS

GenAI & LLMs: OpenAI API, RAG, LangChain, Prompt Engineering, Semantic Search, Agent Workflows

Backend & Programming: Python, SQL, FastAPI, Flask, REST APIs, API Integration

Machine Learning: Scikit-learn, PyTorch, CatBoost, Feature Engineering, Model Evaluation

Data & Cloud: PostgreSQL, pgvector, SQLite, Docker, Google Cloud Run, Git/GitHub

RELEVANT PROJECTS

Cognitive Memory Engine — Python, FastAPI, PostgreSQL, pgvector, RAG, GCP - [Live Demo](#)

- Built a production RAG system on FastAPI, combining pgvector similarity search with LLM generation to return context-grounded answers, and deployed it as a live REST API on Google Cloud Run via Docker.
- Designed, tested, and tuned the embedding and retrieval pipeline end-to-end, iterating on retrieval quality through continuous evaluation.

AI Diary Assistant — Python, Flask, LLMs, Whisper, Notion API - [Live Demo](#)

- Developed a voice-driven LLM assistant with persistent memory, refining prompts to improve multi-turn context retention and response relevance.
- Integrated Whisper speech-to-text with the LLM pipeline and Notion API for structured, searchable knowledge capture.

Food Delivery ETA Prediction — Python, CatBoost, Scikit-learn, Docker - [Live Demo](#)

- Built and deployed an end-to-end ML inference pipeline, benchmarking CatBoost, XGBoost, Random Forest, and LightGBM — CatBoost performed best (R^2 0.84).
- Containerized the prediction service with Docker and deployed it on Google Cloud Run.

RESEARCH & PUBLICATIONS

AI-Driven and NLP-Powered Multilingual Tutor for Indian Languages — Vaishnavi J., Prasanth K. R., Nithish Kumar R., Pragadheesh M., Haresh K. | 8th International Conference on Intelligent Sustainable Systems (ICISS 2026), IEEE-affiliated | Paper ID: ICISS-0103 | Accepted

EDUCATION

M. Kumarasamy College of Engineering — Chennai, Tamil Nadu

Bachelor of Technology in Artificial Intelligence and Data Science | CGPA: 7.2/10 | Graduated 2026

CERTIFICATIONS & ACHIEVEMENTS

- IBM AI Developer Professional Certificate — Generative AI, Prompt Engineering, and LLM-powered application development with Python and Flask.
- NPTEL — The Joy of Computing Using Python (Score: 72%).